

Ericka Bermúdez

Team Lead & Full-stack AI Developer | ericka.dev

Work Experience

Team Lead. *CreAI* — April 2026 – Current

- Lead a cross-functional engineering team of 8 (Full-stack, ML, Data Engineering, Frontend), owning technical direction, individual development plans, and performance tracking across disciplines.
- Remain hands-on as an active contributor: coding features, driving architecture decisions, and conducting code reviews across a 5-service AI document intelligence platform.
- Designed the initial domain data model from scratch — including versioned document lineage, event-sourced audit logging, and multi-tenant schema — later refined collaboratively with ML and graph DB specialists.
- Defined the initial inter-service contracts for an AI microservices pipeline (gRPC, REST, WebSocket), collaborating with architects from high-level design through formal technical review.
- Contributed to a GraphRAG Q&A system using Neo4j and Azure OpenAI, enabling context-aware answers grounded in a structured knowledge graph built from legal documents.
- Coordinate with Product and Delivery stakeholders to define, prioritize, and unblock technical work.

Full-stack AI Developer. *CreAI* — 2025 – April 2026

- Developed applications with Azure Functions, Angular, React, and FastAPI in an AI-driven product environment.
- Collaborated with product managers to define, refine, and prioritize technical requirements, translating business needs into clear technical solutions.
- Optimized an LLM-driven pipeline by introducing response caching and prompt batching, reducing execution time and external API calls by ~50% and lowering operational costs.
- Provided technical guidance to the developer team on implementation patterns, solution structure, and engineering best practices.

Full-stack Developer. *Prograpp* — 2022 – 2025

- Developed core features for Tavaro (formerly Platsage), an omnichannel AI messaging and CRM platform integrating WhatsApp, Instagram, and Facebook for multi-tenant business clients.
- Built a multi-tenant AI agent system where clients configure their own agent personality, knowledge base, and conversation flows using LangChain tools and RAG, with prompt engineering tailored per integration.
- Designed a centralized Meta API middleware handling OAuth authentication and token management across all tenants, allowing clients to connect their Meta channels without individual app registrations.
- Automated tenant infrastructure provisioning using Pulumi; later optimized to per-

tenant DB provisioning to reduce costs.

- Set up AWS middleware infrastructure from scratch using Elastic Beanstalk and Amplify.
- Mentored junior developers on best practices and technology adoption.

Front-End Developer. *X-Labs (Vog App Developers) — 2021 – 2022*

- Developed dynamic and responsive websites using Vue, ReactJS, and Django.
- Utilized CSS frameworks such as Bootstrap and Tailwind CSS to create modern, user-friendly interfaces.

Python Instructor. *IA Center — 2021*

- Delivered a programming seasonal course to over 50 students, teaching foundational Python skills via Microsoft Teams and CISCO platforms.

Full-Stack Developer. *Leading Technologies — 2020 – 2021*

- Designed and implemented data models and API integrations, translating business needs into software solutions.
- Developed e-commerce platforms with Angular, Node.js, MongoDB and Firebase.

Developer Intern. *Universidad Autónoma de Chihuahua — 2019*

Planned, designed, and developed HR department modules using Play Framework (Java), Node.js, and Bootstrap.

Education

Master's Degree on Engineering with Specialty on Data Analytics — 2022 - 2024

Hankuk University of Foreign Studies, South Korea

Bachelor's Degree in Computer Science — 2017-2021

Universidad Autónoma de Chihuahua, Mexico

Bachelor's Degree in Information and Communication Technologies

Management — 2014 - 2017

Universidad Autónoma de Chihuahua, Mexico

Publications

- Activity Transitions for Semi-Supervised Federated Learning in Sensor-Based Human Activity Recognition
- Federated Topology Preserving Domain Adaptation for Human Activity Recognition
- A Framework for Privacy-Preserved Collaborative Learning in Smart Factory Environment

Skills and Certifications

Technical Skills: Languages & Frameworks: Python, TypeScript, FastAPI, Angular, React, Next.js, Vue. AI/ML: Azure OpenAI, LangChain, GraphRAG, Neo4j, RAG pipelines, prompt engineering. Data & Cloud: PostgreSQL, MongoDB/CosmosDB, Azure (Blob Storage, Key Vault, B2C), AWS, Docker, Pulumi. Other: gRPC, REST, WebSocket, Pydantic, SQLAlchemy, Figma, Storybook, Jest, pytest.

Languages: Spanish (native). Fluent English (TOEFL iBT 108). Basic Chinese and Korean.

Certifications & Training:

- Promote the Ethical Use of Data-Driven Technologies (CertNexus)
- People & Soft Skills: Essential for Professional Success Specialization (IBM)